Solution Architecture

Date	06 July 2026
Project Name	**OptiCrop: Smart Agricultural Production Optimization Engine**
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Presentation / Client Layer

Farmer Web Application (HTML, CSS, JavaScript)

↓↑

API Gateway / Load Balancer

[e.g., Flask Web Server]

↓↑

Auth Service	**Core Logic Service**	**External API Integrations**
Request Processing & Input Validation	*Crop Recommendation Model (Random Forest), Prediction Engine, Crop Information Module*	*No External APIs Currently Implemented*

↓↑

Data / Storage Layer

Crop Recommendation Dataset (CSV), Trained Model (model.pkl), Crop Images

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
[Presentation Layer]	[Provides a user-friendly interface for farmers and administrators to access crop recommendations, disease detection, weather updates, and fertilizer suggestions]	[HTML, CSS, JavaScript]
[API Gateway]	[Receives client requests, processes them, and forwards them to the appropriate backend services while handling routing and communication.]	[Flask]
[Microservices]	[Executes machine learning models for crop recommendation, plant disease detection, weather analysis, and fertilizer recommendation.]	[Python, Flask, Scikit-learn, TensorFlow]
[Database]	[Stores farmer information, crop records, prediction results, datasets, and application data securely for future access.]	CSV Dataset, Joblib Model Files